

CHEMISTRY & SUSTAINABILITY

CHEM **SUS** CHEM

ENERGY & MATERIALS

Accepted Article

Title: Photochemical Carboxylation of Activated C(sp³)-H Bonds with CO₂

Authors: Yong-Yuan Gui, Wen-Jun Zhou, Jian-Heng Ye, and Da-Gang Yu

This manuscript has been accepted after peer review and appears as an Accepted Article online prior to editing, proofing, and formal publication of the final Version of Record (VoR). This work is currently citable by using the Digital Object Identifier (DOI) given below. The VoR will be published online in Early View as soon as possible and may be different to this Accepted Article as a result of editing. Readers should obtain the VoR from the journal website shown below when it is published to ensure accuracy of information. The authors are responsible for the content of this Accepted Article.

To be cited as: *ChemSusChem* 10.1002/cssc.201700205

Link to VoR: <http://dx.doi.org/10.1002/cssc.201700205>

WILEY-VCH

www.chemsuschem.org

A Journal of



Photochemical Carboxylation of Activated C(sp³)—H Bonds with CO₂

Yong-Yuan Gui,^{[a]+} Wen-Jun Zhou,^{[a,b]+} Jian-Heng Ye,^[a] and Da-Gang Yu^{*[a]}

As well known, the ever increasing concentration of carbon dioxide (CO₂) in the atmosphere is likely to contribute to global warming. Thus, CO₂ capture and utilization (CCU) has been explored as a potential option in terms of coping with global warming and climate change. However, from the view of organic synthesis, CO₂ may act as an important and ideal one-carbon (C1) source. The utility of CO₂ to provide high-value-added organic chemicals has received considerable attention.^[1] However, it is challenging to utilize CO₂ as a raw material due to its thermodynamic stability and kinetic inertness. As a consequence, direct carboxylation with CO₂ has been mainly confined to highly reactive, and in many instances, air-sensitive organometallic species such as organolithium, Grignard reagents, or organozinc derivatives. The utilization of these organometallic reagents may restrict their general application in organic synthesis, especially the late-stage diversification of densely functionalized molecules. Another powerful alternative to existing methodologies is the development of transition metal-catalyzed reductive carboxylation of organic (pseudo)halides with CO₂ by Martin, Tsuji and Fujihara, and others.^[2] On the other hand, C—H bonds are virtually unlimited in hydrocarbons, natural products, drugs, and synthetic and biological polymers. Direct C—H carboxylation with CO₂ is highly appealing in a greener and more atom-economical way. Many developments have been achieved in the field of C(sp²)—H bonds functionalization with CO₂ by Nolan,^[3] Hou,^[4] Iwasawa,^[5] Olah and Prakash,^[6] our group^[7] and others.^[8] However, carboxylation of more challenging C(sp³)—H bonds with CO₂ achieved much less success, which are mainly limited to the compounds containing acidic C(sp³)—H bonds such as acidic methylene compounds under harsh conditions or high pressure of CO₂.^[9] One formal C(sp³)—H bonds carboxylation was reported by Sato, Mita and coworkers, who realized Ru- or Ir-catalyzed C(sp³)—H bonds silylation and sequential fluoride-mediated carboxylation with CO₂.^[10] Despite the progress achieved, development of direct carboxylation of inert C(sp³)—H bonds under mild conditions is still highly desirable, considering that aliphatic carboxylic acids are one of the most ubiquitous motifs in many compounds and display significant biological activity (Figure 1).^[11]

Over the past decades, light-driven catalysis, especially photoredox catalysis, that harnesses the energy of light to accelerate a chemical reaction, has enabled a variety of nontraditional bond constructions in organic chemistry.^[12] In this context, huge efforts have been devoted to the development of

light-driven CO₂-fixation processes, which mainly focus on the photocatalytic reduction of CO₂ to give carbon monoxide, formic acid, or other reduction products.^[13] In sharp contrast, the light-driven CO₂-fixation via C—C bond formation to give carboxylic acids or its derivatives still remains a relatively unexplored area.

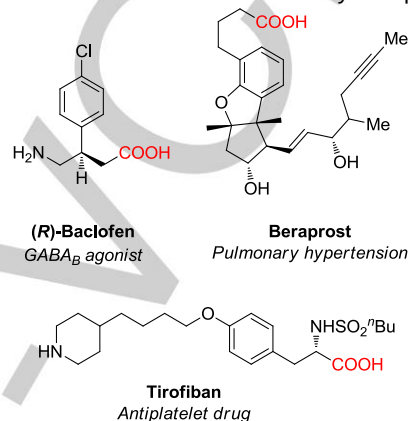
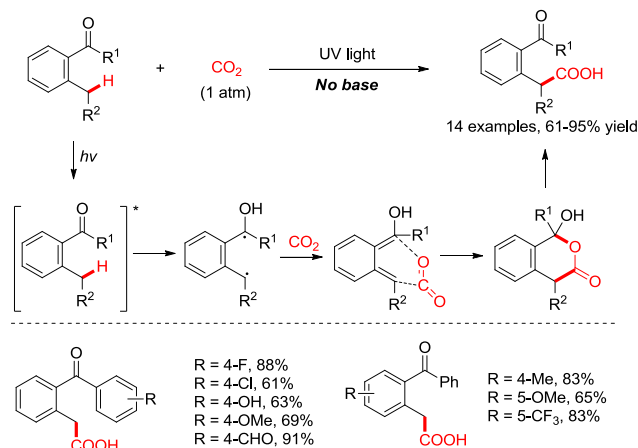


Figure 1 Bioactive carboxylic acids

In late 2015, Murakami and co-workers reported the first example of novel and clean process for direct carboxylation of benzylic C(sp³)—H bonds by simply using UV irradiation or sunlight as driving force (Scheme 1).^[14a] In DMSO solution, *o*-alkylphenyl ketones react efficiently with CO₂ to generate the corresponding *o*-acylphenylacetic acids at room temperature. Mechanistically, *o*-alkylphenyl ketones are promoted to their excited state under photo-irradiation. The excited carbonyl group abstracts a hydrogen at the benzylic position to furnish a 1,4-biradical species, which subsequently generates the *o*-quinodimethane. This 1,3-diene moiety undergoes a [4+2] cycloaddition reaction with the C=O double bond of CO₂ to give a six-membered cycloadduct, which delivers the carboxylation product through ring opening process.^[14] This approach is featured by its remarkable mild reaction conditions, an atmospheric pressure of CO₂, clean process, and no need of base, providing a nice alteration to the traditional thermal carboxylation reaction.



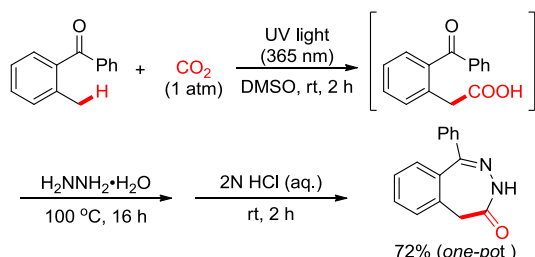
[a] Dr. Y.-Y. Gui,^[1] Prof. Dr. W.-J. Zhou,^[1] J.-H. Ye, Prof. Dr. D.-G. Yu
Key Laboratory of Green Chemistry & Technology of Ministry of Education, College of Chemistry, Sichuan University
29 Wangjiang Road, Chengdu 610064 (P. R. China)
E-mail: dgyu@scu.edu.cn

[b] Prof. Dr. W.-J. Zhou^[1]
College of Chemistry and Chemical Engineering, Neijiang Normal University, Neijiang 641112 (P. R. China)

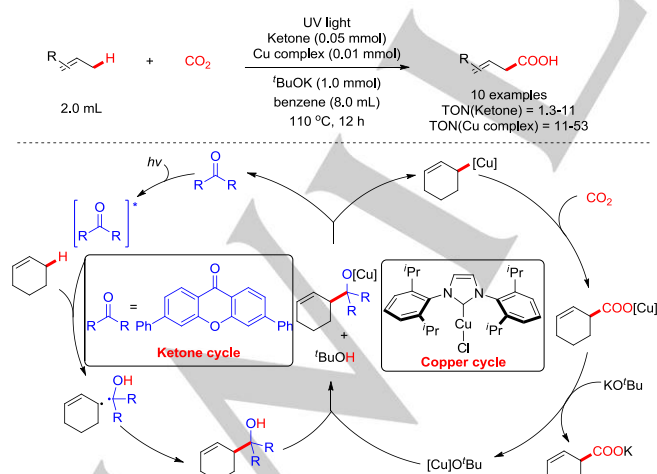
[+]⁺ These authors contributed equally to this work.

Scheme 1 UV light-driven carboxylation of benzylic C(sp³)—H bonds with CO₂

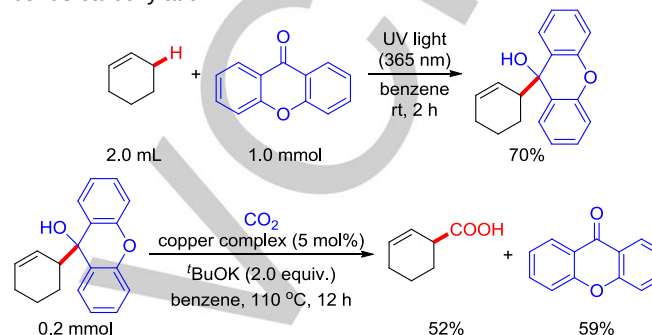
Solar light can also promote such a carboxylation efficiently. When phenyl(*o*-tolyl)methanone was irradiated with solar light in an atmospheric pressure of CO₂ for 7 h on a sunny day, the carboxylic acid was obtained in 72% yield. Moreover, the bioactive 2,3-benzodiazepine could be obtained from easily available starting materials in 72% total yield by using this simple and straightforward carboxylation reaction (Scheme 2).


Scheme 2 Application in synthesis of important pharmacophore-2,3-benzodiazepine

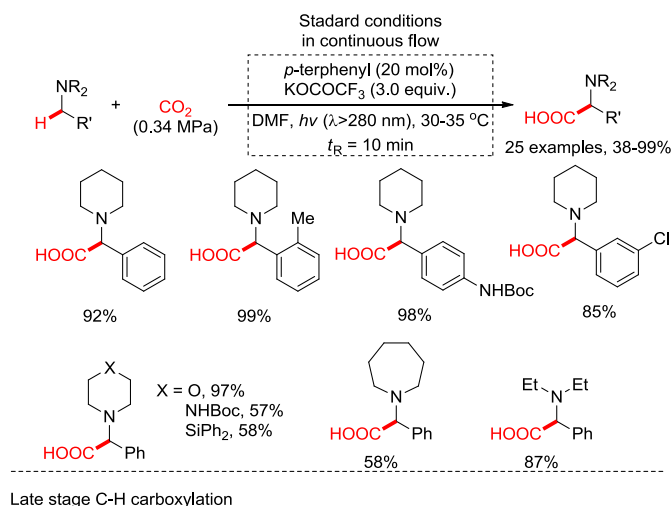
In 2016, Murakami and coworkers further realized the unprecedented carboxylation of allylic C(sp³)—H in simple alkenes with CO₂ (Scheme 3).^[15] After exciting by UV light, the carbonyl oxygen of excited ketone intermolecularly abstracts a hydrogen at the allylic position to generate the radical pair intermediates, which undergo a radical-radical coupling reaction to give tertiary homoallyl alcohol (scheme 3, left, ketone cycle). Then, this energetic alcohol is deprotonated by copper *tert*-butoxide, in-situ generated from copper complex and KO^{*t*}Bu, to give copper alkoxide, which undergoes β-carbon elimination to produce allylcopper species. Then, its nucleophilic addition to CO₂ and further ligand exchange produce the desired product and regenerate the active copper alkoxide catalyst (scheme 3, right, copper cycle). The TON of ketone or copper complex can be up to 11 or 53.


Scheme 3 UV light-promoted allylic C(sp³)—H bonds carboxylation with CO₂

Importantly, the putative intermediates homoallyl alcohol was isolated as a major product (70% isolated yield based on ketone) in the control experiments of cyclohexene and xanthone under standard reaction conditions (Scheme 4). The homoallyl alcohol could generate carboxylic acid in 52% yield and xanthone in 59% yield in the absence of light. These results are consistent with the proposed mechanism and confirm that both ketone and copper complex act as catalysts in such allylic C—H bonds carboxylation.


Scheme 4 Control experiments

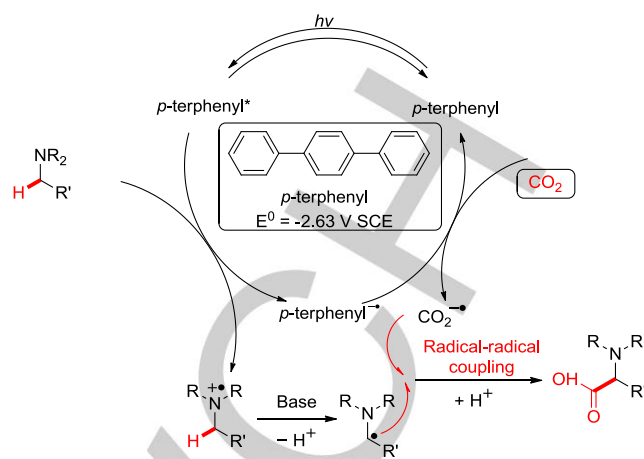
Contrary to the established strategy of photochemical reduction of CO₂ to CO or formic acid using sacrificial reductant (usually a tertiary amine), very recently, Jamison and co-workers disclosed for the first time a photoredox activation strategy of CO₂ and amines for the α-amino acid synthesis in continuous flow (Scheme 5).^[16] One-electron reduction of CO₂ to CO₂^{•-} is highly unfavorable with a formal reduction potential of -2.21 V vs SCE (saturated calomel electrode), which is beyond the photoreducing capabilities of the commonly-used ruthenium and iridium catalysts in photoredox catalysis. Inspired by the work of Yanagida and co-workers who used *p*-terphenyl (*E*⁰ = -2.63 V versus SCE in *N,N*-dimethylformamide) as photoredox catalyst for the reduction of CO₂ to formic acid under ultraviolet light,^[17] Jamison and co-workers applied this organic photocatalyst into their system and successfully realized such a unprecedented process.



Scheme 5 Photoredox α -carboxylation of amines with CO₂ in continuous flow

In this protocol, transformation is operated under continuous flow, which is demonstrated to be more efficient than in batch. Moreover, the reaction is featured by its high regioselectivity, broad substrate scope and mild reaction conditions. In all cases, highly regioselective carboxylations happen at the benzylic positions. The reaction is applicable for a range of *N*-benzylpiperidines bearing different electron-neutral and electron-rich groups on the arenes. Many kinds of functional groups, such as ethers, amides, and ketals, are compatible under the reaction conditions. Beside piperidine, other heterocycles, including piperazine, morpholine, thiomorpholine and silapiperidine, also work nicely to afford the amino acids in good yields. Notably, acyclic amines can also act as efficient coupling partners. The present method also shows its potential application in late stage C–H functionalization of complexes, as exemplified by the regioselective carboxylation of Ticlopidine, a marketed antiplatelet agent.

As shown in Scheme 6, irradiation of the organic catalyst with ultraviolet light produces the excited singlet state of *p*-terphenyl, which is able to oxidize amine to generate the amine radical cation and *p*-terphenyl radical anion. This radical anion is so strongly reducing that it could donate an electron to CO₂ ($E^0 = -2.21$ V vs SCE in DMF) and deliver the corresponding CO₂^{•−}. The radical-radical coupling between α -amino radical and CO₂^{•−} produces α -amino acid as the desired product.



Scheme 6 Proposed mechanism of light-driven carboxylation of amines with CO₂

In order to explore the mechanism, some experimental studies were carried out. Firstly, when 2.5 equivalents of 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) were added under the standard conditions, no desired coupling product was detected, indicating a radical species is involved in the reaction. Furthermore, Stern-Volmer emission quenching experiments were also conducted to demonstrate that the luminescence of the excited state of *p*-terphenyl was quenched by amine instead of CO₂. Lastly, they also calculated the apparent quantum efficiency (AQE) of this transformation to explore the possibility of radical chain. The calculated AQE is 0.045, indicating that radical chain process is unlikely.

Obviously, these pioneering works manifest that exploiting light energy as the driving force has come to be a robust tool for transformations of CO₂. The challenging C(sp³)–H bonds carboxylation has been successfully realized under very mild conditions. Although significant progress has been made by the groups of Murakami and Jamison, there are still many challenges remained to be explored in this emerging field. For example, the types of reactions and substrates with CO₂ need to be significantly expanded, especially for unactivated C(sp³)–H bond. Furthermore, much more attention should be paid to the mechanism of CO₂ activation, which will undoubtedly guide the design of more effective and novel catalytic systems. Last but not least, the asymmetric C(sp³)–H carboxylation should be further developed based on the nice pioneering works. We envision that this highlight will inspire a new direction in the utilization of CO₂ with the assistance of light,^[18] and more challenging work such as direct carboxylation of more normal C(sp³)–H with CO₂ will be realized one day.

Acknowledgements

We thank the National Natural Science Foundation of China (21502124), the “973” Project from the MOST of China (2015CB856600), the “1000-Youth Talents Plan”, and a Start-up Grant from Sichuan University for financial support.

Keywords: photochemistry • carbon dioxide • C(sp³)—H bond functionalization • carboxylation

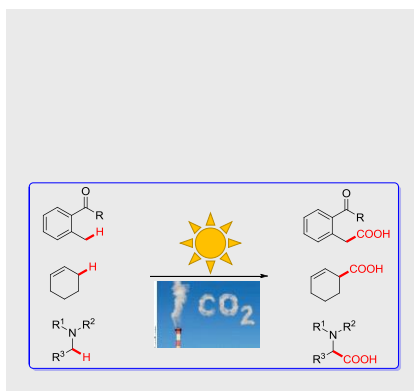
- [1] a) *Carbon Dioxide as Chemical Feedstock* (Ed.: M. Aresta), Wiley-VCH, Weinheim, 2010; b) K. Huang, C.-L. Sun, Z.-J. Shi, *Chem. Soc. Rev.* **2011**, *40*, 2435; c) R. Martin, A. W. Kleij, *ChemSusChem* **2011**, *4*, 1259; d) Y. Tsuji, T. Fujihara, *Chem. Commun.* **2012**, *48*, 9956; e) W. Zhang, X. Lu, *Chin. J. Catal.* **2012**, *33*, 745; f) L. Zhang, Z. Hou, *Chem. Sci.* **2013**, *4*, 3395; g) Q. Liu, L. Wu, R. Jackstell, M. Beller, *Nat. Commun.* **2015**, *6*, 5933; h) B. Yu, L. N. He, *ChemSusChem* **2015**, *8*, 52; i) D. Yu, S. P. Teong, Y. Zhang, *Coord. Chem. Rev.* **2015**, *293–294*, 279; j) S. Wang, G. Du, C. Xi, *Org. Biomol. Chem.* **2016**, *14*, 3666; k) Q. Zhu, L. Wang, C. Xia, C. Liu, *Chin. J. Org. Chem.* **2016**, *36*, 2813; l) G. Yuan, C. Qi, W. Wu, H. Jiang, *Curr. Opin. Green & Sus. Chem.* **2017**, *3*, 22; m) A. W. Kleij, M. North, A. Urakawa, *ChemSusChem* **2017**, DOI: 10.1002/cssc.201700218; n) Z. Zhang, T. Ju, J.-H. Ye, D.-G. Yu, *Synlett* **2017**, DOI: 10.1055/s-0036-1588403.
- [2] For an excellent review, see: a) M. Börjesson, T. Moragas, D. Gallego, R. Martin, *ACS Catal.* **2016**, *6*, 6739; For examples, see: b) A. Correa, R. Martin, *J. Am. Chem. Soc.* **2009**, *131*, 15974; c) T. Fujihara, K. Nogi, T. Xu, J. Terao, Y. Tsuji, *J. Am. Chem. Soc.* **2012**, *134*, 9106; d) T. León, A. Correa, R. Martin, *J. Am. Chem. Soc.* **2013**, *135*, 1221; e) H. Tran-Vu, O. Daugulis, *ACS Catal.* **2013**, *3*, 2417; f) T. Moragas, J. Cornella, R. Martin, *J. Am. Chem. Soc.* **2014**, *136*, 17702; g) Y. Liu, J. Cornella, R. Martin, *J. Am. Chem. Soc.* **2014**, *136*, 11212; h) A. Correa, T. León, R. Martin, *J. Am. Chem. Soc.* **2014**, *136*, 1062; i) K. Nogi, T. Fujihara, J. Terao, Y. Tsuji, *Chem. Commun.* **2014**, *50*, 13052; j) X. Wang, Y. Liu, R. Martin, *J. Am. Chem. Soc.* **2015**, *137*, 6476; k) S. Zhang, W. Q. Chen, A. Yu, L. N. He, *ChemCatChem* **2015**, *7*, 3972; l) K. Nogi, T. Fujihara, J. Terao, Y. Tsuji, *J. Org. Chem.* **2015**, *80*, 11618; m) T. Mita, Y. Higuchi, Y. Sato, *Chem. Eur. J.* **2015**, *21*, 16391; n) T. Moragas, M. Gaydou, R. Martin, *Angew. Chem. Int. Ed.* **2016**, *55*, 5053; o) M. Börjesson, T. Moragas, R. Martin, *J. Am. Chem. Soc.* **2016**, *138*, 7504.
- [3] a) I. I. F. Boogaerts, S. P. Nolan, *J. Am. Chem. Soc.* **2010**, *132*, 8858; b) I. I. F. Boogaerts, G. C. Fortman, M. R. L. Furst, C. S. J. Cazin, S. P. Nolan, *Angew. Chem. Int. Ed.* **2010**, *49*, 8674.
- [4] a) L. Zhang, J. Cheng, T. Ohishi, Z. Hou, *Angew. Chem. Int. Ed.* **2010**, *49*, 8670; b) H. Inomata, K. Ogata, S. Fukuzawa, Z. Hou, *Org. Lett.* **2012**, *14*, 3986.
- [5] a) H. Mizuno, J. Takaya, N. Iwasawa, *J. Am. Chem. Soc.* **2011**, *133*, 1251; b) K. Sasano, J. Takaya, N. Iwasawa, *J. Am. Chem. Soc.* **2013**, *135*, 10954; c) T. Suga, H. Mizuno, J. Takaya, N. Iwasawa, *Chem. Commun.* **2014**, *50*, 14360; d) T. Suga, T. Saitou, J. Takaya, N. Iwasawa, *Chem. Sci.* **2017**, *8*, 1454.
- [6] G. A. Olah, B. Török, J. P. Joschek, I. Bucsí, P. M. Esteves, G. Rasul, G. K. S. Prakash, *J. Am. Chem. Soc.* **2002**, *124*, 11379.
- [7] a) Z. Zhang, L.-L. Liao, S.-S. Yan, L. Wang, Y.-Q. He, J.-H. Ye, J. Li, Y.-G. Zhi, D.-G. Yu, *Angew. Chem. Int. Ed.* **2016**, *55*, 7068; b) Z. Zhang, T. Ju, M. Miao, J.-L. Han, Y.-H. Zhang, X.-Y. Zhu, J.-H. Ye, D.-G. Yu, Y.-G. Zhi, *Org. Lett.* **2017**, *19*, 396.
- [8] a) O. Vechorkin, N. Hirt, X. Hu, *Org. Lett.* **2010**, *12*, 3567; b) K. Nemoto, H. Yoshida, N. Egusa, N. Morohashi, T. Hattori, *J. Org. Chem.* **2010**, *75*, 7855; c) Z. Xin, C. Lescot, S. D. Friis, K. Daasbjerg, T. Skrydstrup, *Angew. Chem. Int. Ed.* **2015**, *54*, 6862; d) W.-Y. Gao, H. Wu, K. Leng, Y. Sun, S. Ma, *Angew. Chem. Int. Ed.* **2016**, *55*, 5472; e) A. Banerjee, G. R. Dick, T. Yoshino, M. W. Kanan, *Nature* **2016**, *531*, 215; f) S. Fenner, L. Ackermann, *Green Chem.* **2016**, *18*, 3804; g) S. Wang, P. Shao, G. Du, C. Xi, *J. Org. Chem.* **2016**, *81*, 6672; h) S. Sun, W.-M. Hu, N. Gu, J. Cheng, *Chem. Eur. J.* **2016**, *22*, 18729; i) J. Luo, S. Preciado, P. Xie, I. Larrosa, *Chem. Eur. J.* **2016**, *22*, 6798.
- [9] a) S. Kikuchi, K. Sekine, T. Ishida, T. Yamada, *Angew. Chem. Int. Ed.* **2012**, *51*, 6989; b) K. Sekine, A. Takayanagi, S. Kikuchi, T. Yamada, *Chem. Commun.* **2013**, *49*, 11320; c) W.-Z. Zhang, M.-W. Yang, X.-B. Lu, *Green Chem.* **2016**, *18*, 4181; d) C.-X. Guo, W.-Z. Zhang, H. Zhou, N. Zhang, X.-B. Lu, *Chem. Eur. J.* **2016**, *22*, 17156. For one example of Lewis acid-promoted carboxylation of allylic C-H bonds, see: e) S. Tanaka, K. Watanabe, Y. Tanaka, T. Hattori, *Org. Lett.* **2016**, *18*, 2576.
- [10] T. Mita, K. Michigami, Y. Sato, *Org. Lett.* **2012**, *14*, 3462.
- [11] *Prodrugs of Carboxylic Acids* (Ed.: H. Maag), Springer, New York, 2007.
- [12] a) C. K. Prier, D. A. Rankic, D. W. C. MacMillan, *Chem. Rev.* **2013**, *113*, 5322; b) Y. Xi, H. Yi, A. Lei, *Org. Biomol. Chem.* **2013**, *11*, 2387; c) D. P. Hari, B. König, *Angew. Chem. Int. Ed.* **2013**, *52*, 4734; d) M. N. Hopkinson, B. Sahoo, J.-L. Li, F. Glorius, *Chem. Eur. J.* **2014**, *20*, 3874; e) D. M. Schultz, T. P. Yoon, *Science* **2014**, *343*, 6174; f) E. Meggers, *Chem. Commun.* **2015**, *51*, 3290; g) J. W. Beatty, C. R. J. Stephenson, *Acc. Chem. Res.* **2015**, *48*, 1474; h) J.-R. Chen, X.-Q. Hu, L.-Q. Lu, W.-J. Xiao, *Chem. Soc. Rev.* **2016**, *45*, 2044; i) J. C. Tellis, C. B. Kelly, D. N. Primer, M. Jouffroy, N. R. Patel, G. A. Molander, *Acc. Chem. Res.* **2016**, *49*, 1429; j) Y.-Y. Gui, L. Sun, Z.-P. Lu, D.-G. Yu, *Org. Chem. Front.* **2016**, *3*, 522.
- [13] H. Takeda, O. Ishitani, *Coord. Chem. Rev.* **2010**, *254*, 346.
- [14] a) Y. Masuda, N. Ishida, M. Murakami, *J. Am. Chem. Soc.* **2015**, *137*, 14063; b) S.-H. Sua, M.-D. Su, *RSC Adv.* **2016**, *6*, 50825.
- [15] N. Ishida, Y. Masuda, S. Uemoto, M. Murakami, *Chem. Eur. J.* **2016**, *22*, 6524.
- [16] H. Seo, M. H. Katcher, T. F. Jamison, *Nat. Chem.* **2016**, DOI:10.1038/nchem.2690.
- [17] S. Matsuoka, T. Kohzaki, C. Pac, A. Ishida, S. Takamuku, M. Kusaba, N. Nakashima, S. Yanagida, *J. Phys. Chem.* **1992**, *96*, 4437.
- [18] Very recently, there have been two elegant reports in photo-driven utilization of CO₂ in organic synthesis, see: a) M.-Y. Wang, Y. Cao, X. Liu, N. Wang, L.-N. He, S.-H. Li, *Green Chem.* **2017**, DOI: 10.1039/c6gc03200a; b) K. Murata, N. Numasawa, K. Shimomaki, J. Takaya, N. Iwasawa, *Chem. Commun.* **2017**, DOI: 10.1039/c7cc00678k

HIGHLIGHT

WILEY-VCH

HIGHLIGHT

Recent great breakthroughs in photochemical carboxylation of the challenging un-acidic C(sp³)—H bonds, including benzylic, allylic and amine C—H bonds, are highlighted here.



Dr. Y.-Y. Gui,^[*] Prof. Dr. W.-J. Zhou,^[*]
J.-H. Ye, Prof. Dr. D.-G. Yu *

Page No. – Page No.

**Photochemical Carboxylation of
Activated C(sp³)—H Bonds with CO₂**